

lateral resolution. The silicon implementation of the beamformer will consume up to 90 per cent less power than an FPGA-based while providing high-end FPGA performance at a lower price, according to Samplify.

Scanning at the speed of flipping pages

A research team at Tokyo University, led by professor Masatoshi Ishikawa, has created a software that can enable users to scan hundreds of pages within minutes.

"It takes a shot of the shape, then it calculates the shape and uses those calculations to film the scanning," said professor Ishikawa. The software is being used in a high-speed camera

plexity, scan times and image quality can all be improved by using two mirrors in the projection head. Its main rationale for using two mirrors is that the vertical and horizontal scan rates differ significantly. For instance, for an 854×480-pixel display, the horizontal scanner must move through 854 steps for each line, while the vertical scanner motion only ticks once. Thus the specified requirements for each mirror are different and, according to bTendo, can be best optimised by manufacturing them separately.

Computer programs to run faster

Researchers at North Carolina State University have found a method to run varied components of some programs, such as wordprocessor and Web browsers, simultaneously. This development is expected to increase the speed of these programs by up to 20 per cent.

The most important unit of a computer chip is its CPU, or 'core'. But to take advantage of these cores, a program has to be broken down into separate threads

to enable each core to work upon a different part of the program simultaneously. The process of simplifying

ing a program into threads is called parallelisation, which makes computers run programs very quickly. The technique invented by North Carolina researchers allows applications to run in parallel.

Low-k dielectrics reduce chip size, but dent life

While the super-insulating low-k dielectric materials help chip designers to reduce chip size, these also shorten the average lifetime of the chip, according to researchers at Columbia University.

Low-k dielectric materials usually have a lot of porosity—vacancy defects—which can act as electron traps, letting current leak through the insulator and eventually short-circuiting the chip.

Usually, engineers evaluate low-k dielectrics by performing a time-dependent dielectric breakdown test, in which they look at the leakage as a function of time, mapping out how long it takes the dielectric to fail but not revealing the underlying mechanism. Columbia University researchers instead measured the average number of density traps in a low-k material, enabling the breakdown mechanism to be identified more easily.

Memristors may replace CPU some day

Memristors (memory resistors) cast in an architecture demonstrate 'stateful logic' that can be dynamically changed between logic operations and memory storage. This configurable architecture demonstrated by Hewlett Packard (HP) can render the dedicated CPU obsolete by enabling dynamically changing circuits to maintain a constant memory of their state.

The architecture makes memristors a candidate for extending



Fastest book scanner

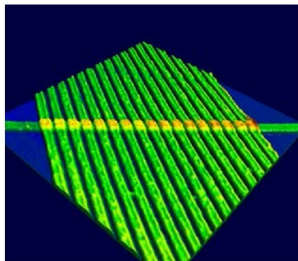
that takes 500 pictures a second to scan pages as they are flipped.

The current system is able to scan an average 200-250-page book in a little over 60 seconds using basic computer hardware that is available off-the-shelf.

Two mirrors make laser scanners perform better

Most laser scanners require only a single microelectromechanical system (MEMS) mirror, which scans images onto the screen, pixel-by-pixel, quite akin to e-beam raster scanning of cathode ray tubes. An Israeli company called bTendo has followed a different approach, adding a second micromirror so that vertical and horizontal scanning are handled separately.

According to bTendo, cost, com-



Memristors